Practical - 4

Aim:
To design set up and configure a Local Area Network (LAN) connecting four computers through a network switch and demonstrate resource/data sharing across the network

Equipment Required

Equipment	Quantity	Specification
personal computer	4	Ethernet(NIC) enabled
Network Switch	1	8/16/24 port
Ethernet cables (cat 5e)	4	straight through, RJ-45 Connectors

IP adress configuration table

Device	IP Adress	Subnet mask	connected to
PC0	10.1.1.1	255.0.0.0	Switch Port Fa0/1
PC1	10.1.1.2	255.0.0.0	Switch Port Fa0/2
PC2	10.1.1.3	255.0.0.0	Switch Port Fa0/3
PC3	10.1.1.4	255.0.0.0	Switch Port Fa0/4
Switch	10.1.1.5	255.0.0.0	VLAN 1 Interface

Implementation in Cisco Packet tracer

1.) Place the Devices
- Open cisco packet tracer
- From the device category panel, Select End Devices and drag 4 PC's onto the workspace.
- Select switches and drag 1 switch.

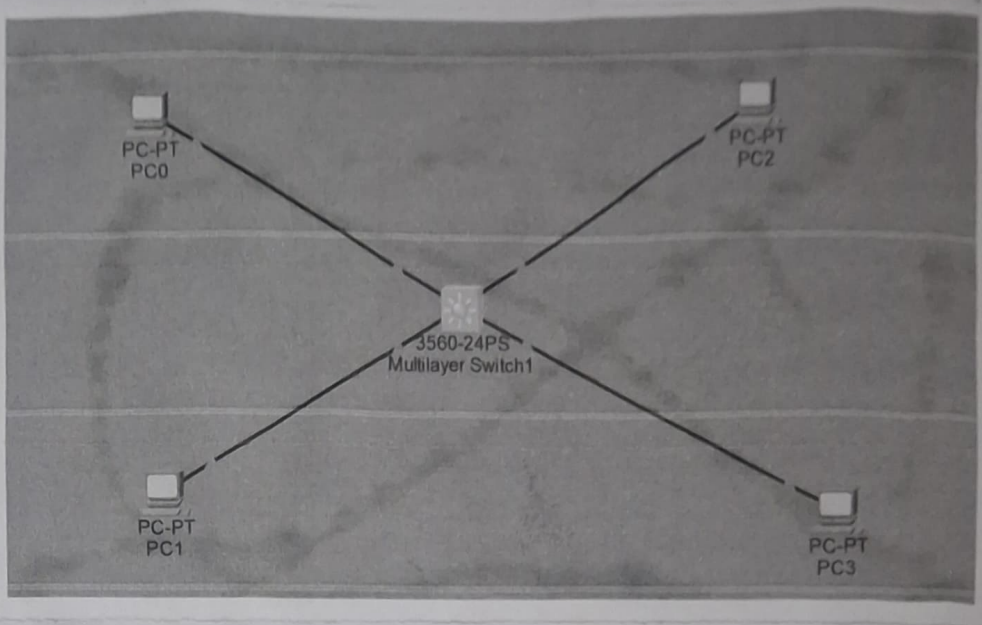
2.) Cable the devices
- click Connections → select Copper straight through cable
- click such PC, then click the switch, and select a available Fast Ethernet port to complete each connection.
- Repeat for all four PCs (ports Fa0/1 to Fa0/4). Green link lights confirm a successful physical connections.

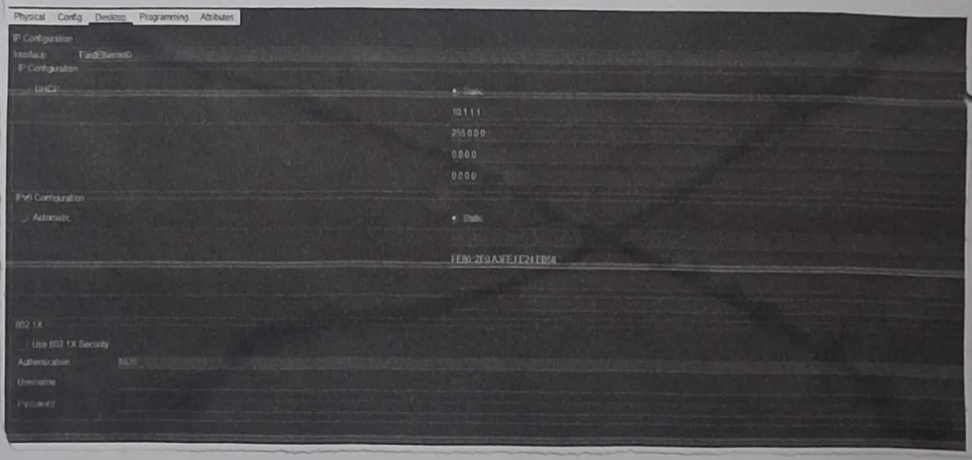
3.) Assign IP Addresses to PC's
- Click a PC → Desktop tab - IP configuration
- Enter the Static IP address and Subnet Mask as per the IP tab

4.) Configure the Switch Management IP
- Click the Switch → CLI tab
- enable
  configure terminal
  interface Vlan 1
  ip address 10.1.1.5 255.0.0.0
  shutdown
  exit

5.) Verify connectivity using Ping
- Click a PC → Desktop tab → cmd prompt
- Use the ping to test connectivity
- successful message

6.) Demonstrate Data Sharing (Server Simulation)
- Add a Server device (End Devices → Server-PT) and connect it to the switch
- Assign it an IP address
- Open the Server's services tab & enable the HTTP
- From another PC, open Web browser enter the server's IP address to load the page, confirming successful data transfer across the LAN.





Result :-

Thus LAN is setted up and connection is verified successfully.